

$$\tau = \dot{m} u_e + (p_e - p_a) A_e$$

Conical Nozzles

Consider the flow out of the conical nozzle of Fig. 11.4, and assume that the streamlines in the expanding part of the nozzle are straight lines that all intersect at the point O . A control surface is indicated, passing through the spherical segment (of radius R) over which exhaust properties are constant. The reaction to the thrust is again shown as the vector $\mathcal{T}$. Applying the momentum equation (2.4), in the axial x -direction gives, for steady flow,

$$\sum F_x = \mathcal{T} + (p_a - p_e) A_e = \int_{cs} \rho (\mathbf{u} \cdot \mathbf{n}) u_x dA,$$

where A_e is the plane exhaust area πr^2 . No mass crosses the control surface except over the spherical exhaust segment, and there it is convenient to let

$$dA = 2\pi R \sin(\phi) R d\phi.$$

differential area in spherical coordinate

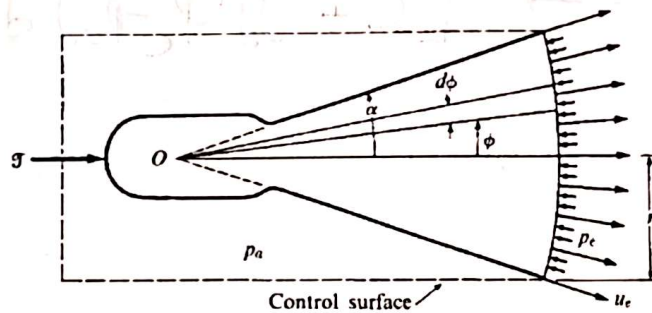
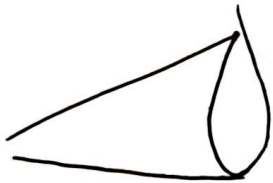
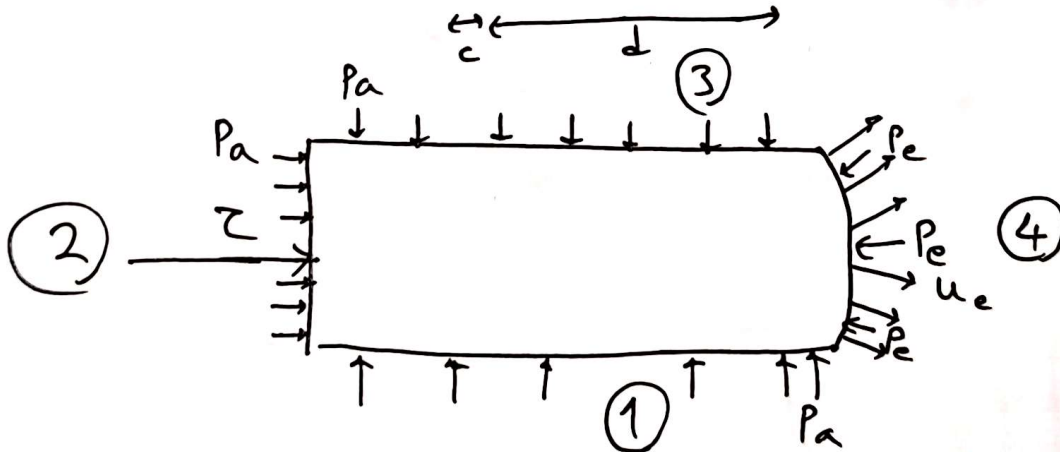


FIGURE 11.4 Spherically symmetric nozzle flow.



$$F = \frac{d}{dt} (mV) = \frac{d}{dt} (\text{Momentum})$$

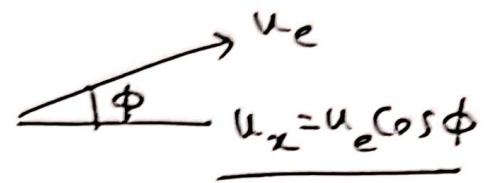
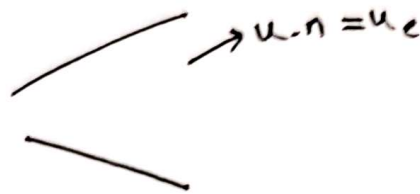
$$X\text{-Mom} \rightarrow \tau + \underbrace{\iint_{(2)} p_a dA + \iint_{(4)} p_e dA}_{\text{I want to get rid of this term}} = \underbrace{\iint_{(4)} (\rho \mathbf{u} \cdot \mathbf{n}) u_x dA}_{\dot{m} u_x}$$

$$\tau + \underbrace{\oint_{(1)(2)(3)} P_a dA}_{\text{surface 1, 2, 3}} + \underbrace{\iint_{(4)} P_e dA}_{\text{surface 4}} + \underbrace{\iint_{(4)} P_a dA}_{\text{surface 4}} - \iint_{(4)} P_a dA = \text{R.H.S}$$

$$\tau + \underbrace{\iint_{(1)(2)(3)} P_a dA}_{\text{surface 1, 2, 3}} + \iint_{(4)} (P_e - P_a) dA = \text{R.H.S}$$

But we knew that $\oint P_a dA = 0$ (over a close surface), so $\Rightarrow$

$$\tau + \iint_{(4)} (P_e - P_a) dA = \text{R.H.S}$$



Further, in this case,

$$\mathbf{u} \cdot \mathbf{n} = u_e \text{ and } u_x = u_e \cos \phi.$$

Thus

$$\mathcal{T} = \iint \rho(u_e) u_e \cos \phi \, dA \quad \text{from previous page}$$

$$\mathcal{T} = \int_0^\pi \rho(u_e) u_e \cos \phi \, 2\pi R^2 \sin \phi \, d\phi + (p_e - p_a) A_e.$$

Integrating, we obtain

$$\textcircled{I} \quad \mathcal{T} = 2\pi R^2 \rho u_e^2 \frac{1 - \cos \alpha}{2} + (p_e - p_a) A_e. \quad \text{use this Eq}^n \text{ for conical nozzles}$$

Since the area of the spherical exhaust segment A'_e is $2\pi R^2(1 - \cos \alpha)$, the propellant flow rate is

$$\dot{m} = \rho u_e A'_e \quad \textcircled{II} \quad \dot{m} = \rho u_e \times 2\pi R^2(1 - \cos \alpha) \quad \text{if you integrate } dA \text{ from previous page}$$

and the thrust equation becomes

$$\text{Separate } \dot{m} \text{ from } \textcircled{II} \text{ in } \textcircled{I}, \text{ we get}$$

$$\mathcal{T} = \dot{m} u_e \frac{1 + \cos \alpha}{2} + (p_e - p_a) A_e \quad \text{Simpler Eq}^n \text{ for conical nozzles (11.13)}$$

or, since the spherical area segment $A'_e = [2/(1 + \cos \alpha)] A_e$,

$$\mathcal{T} = \frac{1 + \cos \alpha}{2} [\dot{m} u_e + (p_e - p_a) A_e].$$

Since, for reasonable nozzles, both the difference between A_e and A'_e and the contribution of the pressure term to the thrust are small, it is adequate to replace A'_e in Eq. (11.13) by the plane area A_e . Thus the thrust of an ideal conical rocket of half-angle α is reduced by a factor λ .

α : nozzle half-angle

divergent loss $\rightarrow \lambda = \frac{1 + \cos \alpha}{2}, \quad (11.14)$

over that of an axial-outlet rocket with the same exhaust conditions.

$$\mathcal{T}_{\text{conical}} = \lambda \mathcal{T}_{\text{uniform}} (A = A'_e)$$

$$A_e = 2\pi R^2 (1 - \cos \alpha)$$

$$A'_e = A_e \left[\frac{2}{1 + \cos \alpha} \right]$$

$$\oint p_a dA = 0$$



α_1 : high λ
Short length
less weight

α_2 : low λ
high length
More weight
 $\alpha_2 < \alpha_1$ but A_2 is the same